

60-minutes of frenzy

Modern Windows Security Bypassed

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EDR / Credential Guard / ASR / Sysmon



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Welcome on stage



Speaker

Sergey Chubarov

MVP Security | Russian Hacker



Speaker

Morten Knudsen

Triple MVP (Security, Azure,
Security Copilot)

Session

Modern Windows Security Bypassed

#wpninjas

Since 2020 we've shown you

And no admin rights required



Defender Antivirus

BYPASSED



**Attack Surface Reduction (via
alternative path)**

BYPASSED



Controlled Folder Access

BYPASSED



Defender for Office 365

BYPASSED



AppLocker

BYPASSED



AMSI in Windows 10 & 11

BYPASSED

Local admin is no longer the master switch

01 / THE CHANGING BASELINE

TAMPER PROTECTION

Security settings resist
ordinary local changes.

PPL

Protected processes
restrict
access and
modification.

VBS

Selected secrets live
beyond the normal OS.

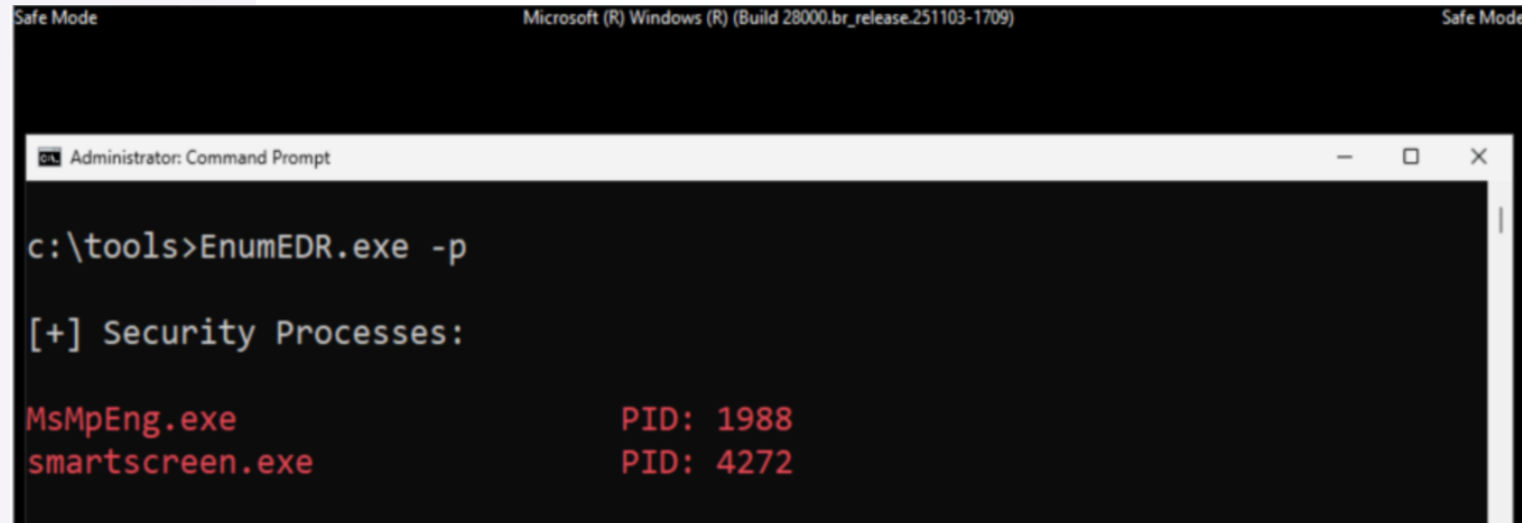
Stronger boundaries change the attacker's route.

Safe Mode ≠ defenses off

01 / SAFE MODE

OLD ASSUMPTION

Minimal boot.
No security controls.



The screenshot shows a Windows desktop in Safe Mode. The taskbar and window title bars indicate 'Safe Mode'. An 'Administrator: Command Prompt' window is open, displaying the output of the command 'c:\tools>EnumEDR.exe -p'. The output lists two security processes: 'MsMpEng.exe' with PID 1988 and 'smartscreen.exe' with PID 4272.

```
Safe Mode Microsoft (R) Windows (R) (Build 28000.br_release.251103-1709) Safe Mode

Administrator: Command Prompt

c:\tools>EnumEDR.exe -p

[+] Security Processes:

MsMpEng.exe PID: 1988
smartscreen.exe PID: 4272
```

When the off-switch fails, change the route

01 / ATTACKER OPTIONS

DISCONNECT

Break the reporting path.

IMPAIR

Abuse privileged components.

GO AROUND

Use exceptions or unprotected assets.

EDR Logic and Evasion Techniques

EDR is a chain-not one process

02 / EDR

SENSOR

Endpoint activity



CLOUD

Behavior + context



RESPONSE

Investigate + contain

Firewall: disconnect the cloud

02 / EDR • DEMO

IMPACT

Telemetry delayed.
Remote response impaired.

HARDENING

Protect network policy:
disable local firewall rules.

Alert on stale endpoints.

Cloud isolation ≠ local protection disabled.

Signed ≠ safe

02 / DRIVER TRUST

PUBLISHER

Develops the driver.

MICROSOFT

Signs new drivers through the hardware program.

CODE INTEGRITY

Checks whether the driver may load.

A trusted signature can belong to a vulnerable driver.

EDR-killing tool

ZEMANA

zam64.sys / zamguard64.sys

Vulnerable anti-malware drivers

Kernel-assisted termination of security processes.

Adlice kernel driver

ADLICE / ROGUEKILLER

TrueSight.sys

Legitimate security software component

Vulnerable versions enable security-process termination.

KsID.sys: Defender as the access path



DEFENDERDUMP / PUBLISHED RESEARCH

Local admin. Existing Microsoft driver. Sensitive memory.

KsID.sys

Microsoft-signed; Defender-supplied

Physical + virtual memory reads

No external driver in the reported scenario

Kernel addresses / KASLR

PPL-protected process memory

Process-token enumeration

Used by KslKatz to dump credentials

RTCore64: a kernel Swiss Army knife



BYOVD / MSI AFTERBURNER / CVE-2019-16098

A signed driver. Multiple paths to kernel abuse.

RTCore64.sys

Arbitrary memory read / write

I/O ports + CPU registers (MSRs)

Privilege escalation

Security-control tampering

Information disclosure

More than process termination: a reusable kernel-access primitive.

Stop the driver before it stops the sensor

02 / BYOVD HARDENING

Driver Signature Enforcement (DSE)

Blocks unsigned or untrusted kernel drivers.



Attack Surface Reduction (ASR)

Blocks writes of known vulnerable signed drivers.



Microsoft vulnerable driver blocklist

Blocks unsafe loads.
Enabled by default:
Windows 11 22H2+.



Memory Integrity (HVCI)

Isolates kernel code integrity using VBS.

Write blocking and load blocking solve different steps.

Secure Boot and Memory Integrity

02 / WINDOWS FOUNDATIONS

SECURE BOOT

Trust the startup chain.

BITLOCKER+TPM

Prevents automatic access to the encrypted drive after boot setting changes

Add driver blocking. Keep firmware and Windows current.

ASR Rule Bypass Techniques

ASR rules block specific behaviors

03 / ATTACK SURFACE REDUCTION

APPLICATION

Who performs the action?



BEHAVIOR

What does the rule match?



DECISION

Block / Audit / Warn

A rule is not a universal execution policy.

Inside the rule: compiled Lua and exceptions

03 / ASR • RULE ANALYSIS

VDM / .LUAC

Compiled rule logic.

PREDICATES

What the rule
evaluates.

EXCEPTIONS

What remains allowed.

Three meanings of "excluded"

03 / EXCLUSIONS

BUILT-IN

Compatibility logic
inside the product.

ASR POLICY

Exceptions configured
by the organization.

AV POLICY

Antivirus exclusions
with different
semantics.

Different exceptions. Different mitigations.

Close the path the rule leaves open

03 / ASR HARDENING

NARROW EXCEPTIONS

Scope. Owner. Expiry.

CONSTRAIN EXECUTION

App Control.
Only required
capabilities.

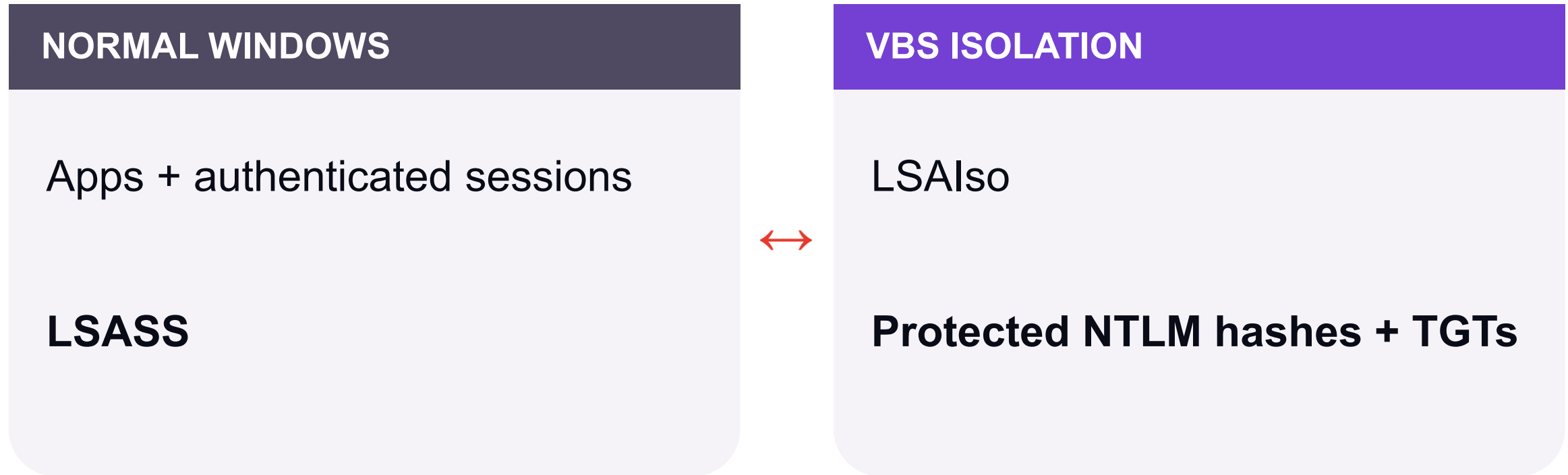
VERIFY THE ACTION

Test actual blocking.
Audit is not prevention.

Testing Credential Guard Protections

Protect the secret-not every use of identity

04 / CREDENTIAL GUARD



Authentication can work without exposing the protected secret.

PPL ≠ Credential Guard

04 / TWO DIFFERENT PROTECTIONS

LSASS / PPL

Restricts process access.

CREDENTIAL GUARD

Isolates selected secrets.

Removing PPL does not automatically disable LSASS.

Credential exposure outside LSAlso

04 / CREDENTIALS • DEMO

NORMAL-OS PATH

LSASS protection.
WDigest credential handling.

ISOLATED STORE

Credential Guard can remain
a separate boundary.

Reduce exposure around the isolated store

04 / CREDENTIAL HARDENING

PROTECT THE PLATFORM

LSA protection. VBS.
Restrict debug access.

SEPARATE ADMIN

Keep privileged
sessions
off everyday endpoints.

FIX THE AUTH PATH

Reduce legacy
credentials.
Revoke exposed
sessions.

Sysmon Evasion and Tampering

Sysmon records. Your pipeline detects.

05 / SYSMON

CAPTURE

Service + driver



RECORD

Local event log



FORWARD

Collector / SIEM

Attack capture-or attack delivery

05 / SYSMON RISK

CAPTURE

Driver impairment.
Configuration gaps.

DELIVERY

Forwarding interrupted.
Events never reach the
SOC.

A startup change can become a logging gap

05 / SYSMON • DEMO

BEFORE REBOOT

Startup config changes.
Driver still loaded.

AFTER REBOOT

Driver may not start.
Coverage can disappear.

Watch configuration changes-and the next boot.

Preserve evidence. Detect silence.

05 / SYSMON HARDENING

PROTECT CONFIG

Restrict changes.
Track drift.

FORWARD EARLY

Keep protected
off-host records.

MONITOR HEALTH

Freshness. Missing
events.
Post-reboot coverage.

Closing the gaps

Remove the options attackers do not need

06 / REDUCE THE ATTACK SURFACE

OLD DRIVERS

Retire unsafe versions
and privileged utilities.

UNUSED FEATURES

Macros. Script hosts.
Unnecessary remote
access.

EXCESS AUTHORITY

Local admin. Debug
access.
Persistent exceptions.

Match the hardening to the failure

06 / CLOSING THE GAPS

Cloud disconnected → Protect policy + detect stale data

Unsafe driver admitted → Driver blocking + HVCI + patching

Credentials exposed → Isolation + session / auth hardening

Telemetry lost → Protected config + off-host evidence

Harder to disable. Still possible to go around.

06 / TAKEAWAYS

PROTECT TRUST

Boot. Kernel.
Security-policy
authority.

REDUCE OPTIONS

Privilege. Features.
Exceptions.

PROVE COVERAGE

Enforcement.
Freshness.
Response completion.

Session materials

RESOURCES

GITHUB REPOSITORY

wpninja2026

github.com/schubarov/wpninja2026

■ Slides

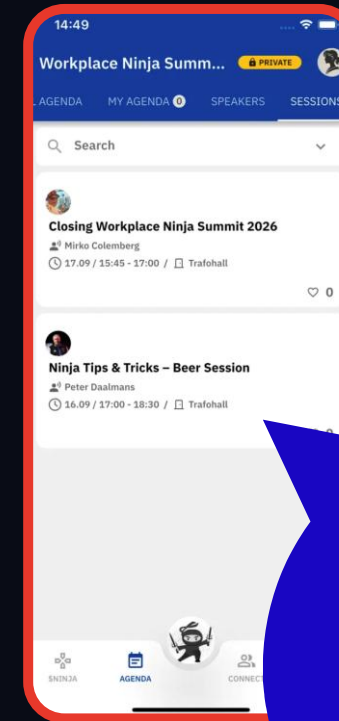
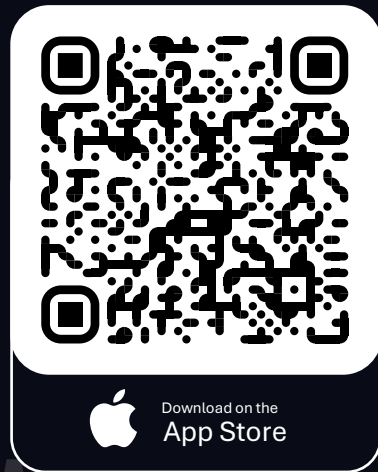
■ ASR exclusions + Lua bytecode

■ Tools



SCAN TO OPEN ON GITHUB

Check in and rate this session in the Workplace Ninja Summit 2026 app



We appreciate your feedback!

Thank You, Ninjas!



Appendix: Additional Attacks & Mitigations

A running process can still be an impaired sensor

APPENDIX / ADDITIONAL EDR EVASION

FAILURE

Unresponsive processing.
Broken cloud delivery.

DEFENSE

Check real data freshness.
Verify completed actions.

Use the session-not the protected secret

APPENDIX / CREDENTIAL SCOPE

OUTSIDE THE STORE

Application secrets.
Existing identity authority.

REDUCE EXPOSURE

Separate admin sessions.
Revoke exposed access.

VBS rollback: trusted does not mean current

APPENDIX / HISTORICAL VBS CASE

HISTORICAL ISSUE

Older components could reintroduce fixed weaknesses.

MITIGATION

Current servicing.
Applicable anti-rollback policy.

A different behavior may need a different rule

APPENDIX / ASR SCOPE

ONE PREDICATE

Office child-process creation.

OTHER BEHAVIORS

Executable content.
Injection. Macro API use.

The event was never selected

APPENDIX / SYSMON COVERAGE

COVERAGE GAP

Missing event type.
Overbroad filter.

DEFENSE

Map detections to fields.
Test benign markers.

Sources: EDR and Windows hardening

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